# ERASURE ENTROPIES OF SPARSE LINEAR DNFS

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Abstract. For a Boolean function $f$ and a coordinate $i$, the erasure entropy $E_i(f)$ is the conditional entropy of the event $i \in U$ given $U \setminus \{i\}$, where $U$ is drawn from the spectral distribution $\hat{f}^2$. The Fourier entropy-influence conjecture is equivalent to every non-constant $f$ having a coordinate with $E_i(f) = O(\mathrm{Inf}_i(f))$. We consider monotone DNFs whose terms all have width $w$, any two of which share at most one variable, and in which every variable lies in at most $D$ terms. We show that $E_i(f)/\mathrm{Inf}_i(f) = 2 + o(1)$ at every relevant coordinate, uniformly in the number of variables and terms, as $w \to \infty$ with $D \le 2^{(1-\eta)(w-1)}$ for a fixed $\eta > 0$, and we give an explicit error bound for each $w$, $D$ and degree of $i$. The proof rests on a separation inequality valid for every monotone function: if the derivative $D_i f$ puts little spectral weight outside a family $\Omega$ of sets and the restriction $f|_{x_i=0}$ puts little weight on $\Omega$, then $E_i/\mathrm{Inf}_i$ is close to 2. For a linear DNF, $\Omega$ is the family of sets that contain many of the other variables of some term through $i$.

## 1. Introduction

For $f\colon \{0,1\}^n \to \{-1,1\}$ write $f = \sum_S \hat{f}(S)\chi_S$ with $\chi_S(x) = (-1)^{\sum_{j\in S} x_j}$, and let $U$ be a random subset with $\Pr[U = S] = \hat{f}(S)^2$. The Fourier entropy is $H(f) = H(U)$ and the total influence is $I(f) = \mathbb{E}|U| = \sum_i \mathrm{Inf}_i(f)$, where $\mathrm{Inf}_i(f) = \Pr[i \in U]$. The Fourier entropy-influence (FEI) conjecture of Friedgut and Kalai [3] asks for an absolute $C$ with $H(f) \le C\, I(f)$ for every Boolean $f$. It is known for symmetric functions [10] and for read-once formulas, over gates of bounded arity [9] and over AND and OR gates [2]. It was proved for read-$k$ decision trees with $k$ fixed in [15] and for linear threshold functions in [13]. Its min-entropy version is known for read-$k$ DNFs with $k$ fixed [11, 1], and Han [5] bounds $H(f)$ by a multiple of $I(f)$ plus $\sum_i \mathrm{Inf}_i(f)\log_2(1/\mathrm{Inf}_i(f))$, with the multiple lowered to the sharp value $1/\ln 2$ by Li and Han [8]. Kelman, Kindler, Lifshitz, Minzer and Safra [7] prove a weaker form of the conjecture: the spectrum of a Boolean function of constant variance is concentrated on $2^{O(I(f)\log I(f))}$ characters.

The erasure entropy (in the sense of Verdú and Weissman [14]) of $f$ at $i$ is

$$E_i(f) = H\big(\mathbf{1}[i \in U] \,\big|\, U \setminus \{i\}\big).$$

It is the part of the entropy drop from $f$ to its two restrictions at $x_i$ that is not controlled by a constant times $\mathrm{Inf}_i$ [5, Remarks 3.1–3.2], [4, (32)], and the FEI conjecture holds if and only if there is an absolute $c$ such that every non-constant Boolean function has a coordinate with $E_i(f) \le c\,\mathrm{Inf}_i(f)$ [12]. The trivial bound $E_i \le h_2(\mathrm{Inf}_i)$ loses a factor $\log(1/\mathrm{Inf}_i)$; the question is when that factor can be removed, and at which coordinates.

This paper computes the ratio $E_i/\mathrm{Inf}_i$ asymptotically for a class of DNFs in which the coordinates are linked by many terms. Let $F$ be a monotone DNF on $\{0,1\}^n$. We call $F$ *linear* if any two of its terms share at most one variable, *$w$-uniform* if all its terms have $w$ variables, and of *degree at most $D$* if every variable lies in at most $D$ terms. Put $f = 1 - 2F$. Read-once

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DNFs are the linear DNFs of degree one; a linear DNF of degree $D$ is read-$D$, but its terms may overlap in a pattern with many cycles, for instance the $m$ rows and $m$ columns of an $m \times m$ array.

**Theorem 1.** *Fix $\eta \in (0,1)$. Over all $n$, all $w$-uniform linear monotone DNFs $F$ on $\{0,1\}^n$ of degree at most $D \le 2^{(1-\eta)(w-1)}$, and all coordinates $i$ with $\mathrm{Inf}_i(f) > 0$,*

$$\sup\Big|\frac{E_i(f)}{\mathrm{Inf}_i(f)} - 2\Big| \longrightarrow 0 \qquad (w \to \infty).$$

*In particular $\sum_i E_i(f) = (2 + o(1))\, I(f)$ on this class.*

Theorem 1 follows from an explicit inequality, Theorem 6, valid for each width, degree and coordinate. The value 2 is the limit of $E_i/\mathrm{Inf}_i$ for a coordinate inside one long AND that nothing else touches: for an AND of $k+1$ variables a direct computation gives $E_i/\mathrm{Inf}_i = 2(1-2^{-k}) + 2^k \Phi\big((1-2^{-k})^2, 4^{-k}\big)$ at each of them (with $\Phi$ as in Section 2), which differs from 2 by at most $2^{-k}(2k+2)$. Theorem 1 says that in a sparse linear DNF every coordinate behaves like that, however many terms pass through it and however the terms around it overlap.

The proof separates two spectra. Write $g$ and $h$ for the even and odd parts of $f$ in $x_i$. For a monotone $F$, $h$ is the indicator that $i$ is pivotal and $g = 2N - 1 - h$, where $N$ is the indicator that $F$ vanishes at $x_i = 0$. If the spectrum of $h$ lives on a family $\Omega$ of sets on which $N$ has almost no weight, then on $\Omega$ the two coefficients $\hat g(T)$ and $\hat h(T)$ have nearly the same square, and each such $T$ contributes nearly $2\hat h(T)^2$ to $E_i$. Proposition 2 makes this quantitative for every monotone function. For a linear DNF the family is $\Omega_k$, the sets that contain at least $k$ of the other variables of some term through $i$: the pivotal indicator spreads its spectral weight evenly over the subsets of the other variables of each such term (Lemma 4), while the background event $N$ needs $k$ further terms to reach $k$ of these variables, which costs $((D-1)2^{-(w-1)})^k$ (Lemma 5).

Theorem 1 is a statement about the ratio at every coordinate and does not prove the FEI conjecture for the class. The equivalence of [12] needs a good coordinate in every restriction of $f$, and restrictions of a uniform linear DNF have terms of different widths.

## 2. A separation inequality

Fix $i$ and write $f^{(b)}$ for the restriction $x_i = b$, a function of the other coordinates. Put

$$g = \tfrac12\big(f^{(0)} + f^{(1)}\big), \qquad h = \tfrac12\big(f^{(0)} - f^{(1)}\big).$$

Then $\hat g(T) = \hat f(T)$ and $\hat h(T) = \hat f(T \cup \{i\})$ for $T \subseteq [n] \setminus \{i\}$, and $\mathbb{E}h^2 = \mathrm{Inf}_i(f) =: \delta$. Put $p(T) = \hat g(T)^2$ and $q(T) = \hat h(T)^2$, and

$$\Phi(a,b) = \ell(a) + \ell(b) - \ell(a+b), \qquad \ell(x) = -x\log_2 x.$$

Since $(U \setminus \{i\}, \mathbf{1}[i \in U])$ takes the value $(T,0)$ with probability $p(T)$ and $(T,1)$ with probability $q(T)$, we have $E_i(f) = \sum_T \Phi(p(T), q(T))$. The function $\Phi$ is concave, homogeneous of degree one, increasing in each argument, and $\Phi(b,b) = 2b$. Writing $\Phi(a,b) = a\log_2(1 + b/a) + b\log_2(1 + a/b)$ and using $\ln(1+t) \le t$ gives

$$\Phi(a,b) \le b\log_2\Big(e\,\frac{a+b}{b}\Big). \tag{1}$$

Now let $F$ be monotone increasing and $f = 1 - 2F$. Put $N = (1+f^{(0)})/2$ and $M = (1+f^{(1)})/2$, the indicators that $F$ vanishes at $x_i = 0$ and at $x_i = 1$. Then $M \le N$, $h = N - M$ is the indicator that $i$ is pivotal, $\hat h(\emptyset) = \mathbb{E}h = \delta$, and

$$g = N + M - 1 = 2N - 1 - h, \qquad \hat g(T) = 2\hat N(T) - \hat h(T) \quad (T \neq \emptyset).$$

**Proposition 2.** *Let $F$ be monotone, $f = 1 - 2F$, $\delta = \mathrm{Inf}_i(f) > 0$, and let $N$ be as above. Let $\Omega$ be a family of non-empty subsets of $[n] \setminus \{i\}$ and let $\varepsilon \in (0, 1]$ and $\kappa \geq 0$ satisfy*

$$\sum_{T \notin \Omega} \hat{h}(T)^2 \leq \varepsilon\delta, \qquad \sum_{T \in \Omega} \hat{N}(T)^2 \leq \kappa\delta.$$

*Then*

$$2 - 2\varepsilon - \frac{8}{\ln 2}\sqrt{\kappa} \;\leq\; \frac{E_i(f)}{\delta} \;\leq\; 2 + \frac{8}{\ln 2}\sqrt{\kappa} + \varepsilon \log_2\Big(e\Big[3 + \frac{8\mathrm{Var}N}{\varepsilon\delta}\Big]\Big) + \delta \log_2\big(e[1 + \delta^{-2}]\big).$$

*Proof.* Split the sum $E_i = \sum_T \Phi(p(T), q(T))$ into $T \in \Omega$, $T \notin \Omega$ with $T \neq \emptyset$, and $T = \emptyset$.

On $\Omega$. Fix $b \geq 0$. The function $u \mapsto \Phi(u^2, b^2)$ is even, equals $2b^2$ at $u = \pm b$, and for $u \neq 0$ has derivative $2u \log_2(1 + b^2/u^2)$, whose absolute value is at most $(4/\ln 2)b$ because $\ln(1 + t^2) \leq 2t$ for $t \geq 0$. For $T \in \Omega$ take $b = |\hat{h}(T)|$ and move $u$ from $-\hat{h}(T)$ to $\hat{g}(T) = 2\hat{N}(T) - \hat{h}(T)$, a distance $2|\hat{N}(T)|$:

$$\big|\Phi(p(T), q(T)) - 2q(T)\big| \leq \frac{8}{\ln 2}|\hat{h}(T)||\hat{N}(T)|.$$

By Cauchy–Schwarz, $\sum_\Omega |\hat{h}||\hat{N}| \leq \sqrt{\delta}\sqrt{\kappa\delta}$, and $(1 - \varepsilon)\delta \leq \sum_\Omega q \leq \delta$. Hence

$$2(1 - \varepsilon)\delta - \frac{8}{\ln 2}\delta\sqrt{\kappa} \leq \sum_{T \in \Omega} \Phi(p(T), q(T)) \leq 2\delta + \frac{8}{\ln 2}\delta\sqrt{\kappa}.$$

Since every term of $E_i$ is non-negative, this gives the lower bound.

Off $\Omega$. Let $P$ and $Q$ be the sums of $p$ and $q$ over the non-empty $T \notin \Omega$; if $Q = 0$ these terms vanish, so let $Q > 0$. Then $Q \leq \varepsilon\delta$, and from $p = (2\hat{N} - \hat{h})^2 \leq 8\hat{N}^2 + 2\hat{h}^2$ we get $P \leq 8\mathrm{Var}N + 2\varepsilon\delta$. By (1) and Jensen's inequality for the concave function $t \mapsto \log_2 t$,

$$\sum \Phi(p, q) \leq \sum q \log_2\Big(e\,\frac{p+q}{q}\Big) \leq Q \log_2\Big(e\,\frac{P+Q}{Q}\Big) \leq Q \log_2\Big(e\,\frac{c}{Q}\Big), \qquad c = 8\mathrm{Var}N + 3\varepsilon\delta.$$

The last expression increases in $Q$ on $(0, c]$, so it is at most $\varepsilon\delta \log_2(e[3 + 8\mathrm{Var}N/(\varepsilon\delta)])$.

At $T = \emptyset$, $q = \delta^2$ and $p \leq 1$, so (1) gives $\Phi \leq \delta^2 \log_2(e[1 + \delta^{-2}])$. Adding the three parts gives the upper bound. □

## 3. Linear DNFs

From now on $F$ is a $w$-uniform linear monotone DNF of degree at most $D$, $f = 1 - 2F$, $L = w - 1 \geq 1$ and $a = 2^{-L}$. Fix a coordinate $i$ that lies in $d \geq 1$ terms $t_1, \ldots, t_d$. The *mates* of $t_s$ are its $L$ variables other than $i$, and $A_s$ is the indicator that all of them equal 1. By linearity the mate sets are pairwise disjoint. A *background term* is a term that does not contain $i$; $N$ is the indicator that every background term fails, and

$$M = N \prod_{s=1}^{d} (1 - A_s), \qquad h = N - M = N \cdot \mathbf{1}\big[\textstyle\bigvee_s A_s\big], \qquad \delta = \mathbb{E}h.$$

Put $A = \mathbb{E}N$.

Two facts about linear DNFs are used repeatedly. A background term meets the mate set of $t_s$ in at most one variable, since it shares at most one variable with $t_s$ and does not contain $i$. And a term through a mate of $t_s$, other than $t_s$ itself, is a background term, since otherwise it would share two variables with $t_s$. We also use Harris's inequality [6]: under a product measure, an increasing and a decreasing event are negatively correlated, and two increasing events are positively correlated.

**Lemma 3.** *With $\theta = 1 - L(D - 1)a - (d - 1)a/2$,*

$$da\,\theta\,A \leq \delta \leq da\,A.$$

*Proof.* The events $A_s$ are increasing and $N$ is decreasing, so $\Pr[N, A_s] \le Aa$, and the union bound gives $\delta \le daA$. For the lower bound, let $R_s$ be the indicator that every background term disjoint from the mate set of $t_s$ fails, so that $N \le R_s$. The remaining background terms meet that mate set in exactly one variable; each mate lies in at most $D-1$ of them, so there are at most $L(D-1)$, and given $A_s = 1$ each of them fires exactly when its other $L$ variables, which are not mates of $t_s$, all equal 1. Let $V_s$ be the event that one of these residual conjunctions equals 1; it is increasing and $\Pr[V_s] \le L(D-1)a$. Since $R_s$ and $V_s$ do not depend on the mates of $t_s$,

$$\Pr[N, A_s] = a \Pr[R_s, \overline{V_s}] \ge a\big(\Pr[R_s] - \Pr[R_s]\Pr[V_s]\big) \ge aA\big(1 - L(D-1)a\big),$$

using Harris for $R_s$ and $V_s$ and, in the last step, $\Pr[R_s] \ge A$; if $1 - L(D-1)a < 0$ the bound holds trivially. For $s \ne u$ the event $A_sA_u$ is increasing with probability $a^2$, so $\Pr[N, A_s, A_u] \le Aa^2$. Bonferroni's inequality gives $\delta \ge daA(1 - L(D-1)a) - \binom{d}{2}a^2A = da\theta A$. □

For $1 \le k \le L$ let $\Omega_k$ be the family of sets $T \subseteq [n] \setminus \{i\}$ that contain at least $k$ mates of some $t_s$, and put

$$\gamma_k = a \sum_{j<k} \binom{L}{j}.$$

**Lemma 4.** $\sum_{T \notin \Omega_k} \hat{h}(T)^2 \le \min(1, d\gamma_k)\, \delta$.

*Proof.* Put $h_s = NA_s \prod_{u<s}(1 - A_u)$. These functions have disjoint supports and sum to $h$, so $\sum_s \|h_s\|_2^2 = \delta$. Since $h_s$ vanishes unless $A_s = 1$, $h_s = A_s\Psi_s$, where $\Psi_s$ is $N\prod_{u<s}(1-A_u)$ with the mates of $t_s$ set to 1, a function of the variables outside that mate set. As $A_s = 2^{-L}\sum_{S \subseteq \mathrm{mates}(t_s)} (-1)^{|S|}\chi_S$, every Fourier coefficient of $h_s$ has the form $\hat{h}_s(S \cup V) = \pm 2^{-L}\hat{\Psi}_s(V)$ with $S$ a set of mates and $V$ disjoint from them. The weight of $\hat{h}_s^2$ on sets with exactly $j$ mates of $t_s$ is therefore $\binom{L}{j}4^{-L}\|\Psi_s\|_2^2 = \binom{L}{j}a\|h_s\|_2^2$, using $\|h_s\|_2^2 = a\|\Psi_s\|_2^2$. A set outside $\Omega_k$ has fewer than $k$ mates of $t_s$, so $\sum_{T \notin \Omega_k} \hat{h}_s(T)^2 \le \gamma_k \|h_s\|_2^2$. By Cauchy–Schwarz,

$$\sum_{T \notin \Omega_k} \hat{h}(T)^2 \le d \sum_s \sum_{T \notin \Omega_k} \hat{h}_s(T)^2 \le d\gamma_k \delta.$$

The bound by $\delta$ is trivial. □

**Lemma 5.** *If $k(D-1)a < 2$, then*

$$\sum_{T \in \Omega_k} \hat{N}(T)^2 \le d\,\alpha_k A, \qquad \alpha_k = \binom{L}{k} \frac{2^{\binom{k}{2} - 2k}\big((D-1)a\big)^k}{1 - k(D-1)a/2}.$$

*Proof.* Every $T \in \Omega_k$ contains a set $J$ of $k$ mates of some $t_s$, and there are $d\binom{L}{k}$ such sets $J$. For each of them, $\sum_{T \supseteq J} \hat{N}(T)^2 = \|D_J N\|_2^2$, where $D_J$ is the composition of the derivatives $D_j N = \frac{1}{2}(N|_{x_j=0} - N|_{x_j=1})$ over $j \in J$. So it suffices to show $\|D_J N\|_2^2 \le \alpha_k \binom{L}{k}^{-1} A$.

Fix $J$. A background term contains at most one element of $J$. Let $R'$ be the indicator that every background term avoiding $J$ fails, and for $j \in J$ let $Q_j$ be the OR of the residual conjunctions, of width $L$, of the at most $D-1$ background terms through $j$. Neither $R'$ nor any $Q_j$ depends on the variables in $J$, and

$$N = R' \prod_{j \in J}(1 - x_j Q_j), \qquad D_J N = \pm R' \prod_{j \in J} \tfrac{1}{2} Q_j, \qquad \|D_J N\|_2^2 = 4^{-k} \Pr[R', \textstyle\bigwedge_j Q_j].$$

The event $\bigwedge_j Q_j$ is increasing and $R'$ decreasing, so $\Pr[R', \bigwedge_j Q_j] \le \Pr[R']\Pr[\bigwedge_j Q_j]$. For $\bigwedge_j Q_j$ one residual conjunction must equal 1 for each $j$; there are at most $(D-1)^k$ choices,

the chosen terms are distinct because each contains exactly one element of $J$, and two residual conjunctions share at most one variable, so their union has at least $kL - \binom{k}{2}$ variables. Hence $\Pr[\bigwedge_j Q_j] \le (D-1)^k 2^{\binom{k}{2}} a^k$. Finally, $x_j$ is independent of $R'$ and $Q_j$, so by Harris

$$A \ge \Pr[R'] - \sum_{j\in J} \Pr[R', x_j Q_j] \ge \Pr[R']\Big(1 - \sum_j \tfrac{1}{2}\Pr[Q_j]\Big) \ge \Pr[R']\big(1 - \tfrac{1}{2}k(D-1)a\big).$$

Combining the three displays gives $\|D_J N\|_2^2 \le 4^{-k}(D-1)^k 2^{\binom{k}{2}} a^k A/(1-k(D-1)a/2)$. □

**Theorem 6.** *Let $F$ be a $w$-uniform linear monotone DNF of degree at most $D$, $L = w-1$, $a = 2^{-L}$, and let $i$ lie in $d \ge 1$ terms. Let $1 \le k \le L$, and put*

$$\theta = 1 - L(D-1)a - \frac{(d-1)a}{2}, \qquad \varepsilon = \min(1, d\gamma_k), \qquad \kappa = \frac{\alpha_k}{a\theta},$$

*with $\gamma_k$ and $\alpha_k$ as above. If $\theta > 0$, $k(D-1)a < 2$ and $da \le 1/4$, then*

$$\Big|\frac{E_i(f)}{\mathrm{Inf}_i(f)} - 2\Big| \le 2\varepsilon + \frac{8}{\ln 2}\sqrt{\kappa} + \varepsilon \log_2\Big(e\Big[3 + \frac{8}{\varepsilon\, da\, \theta}\Big]\Big) + da \log_2\big(e[1 + (da)^{-2}]\big).$$

*Proof.* Apply Proposition 2 with $\Omega = \Omega_k$, which consists of non-empty sets. Lemma 4 gives the first hypothesis with $\varepsilon$ (note $\varepsilon \ge da > 0$). Lemmas 5 and 3 give $\sum_{\Omega_k} \hat{N}^2 \le d\alpha_k A \le \alpha_k \delta/(a\theta) = \kappa\delta$. In the upper bound, $\mathrm{Var} N \le \mathbb{E}N = A \le \delta/(da\theta)$, and $\delta \le daA \le da \le 1/4$; the function $x \mapsto x\log_2(e[1+x^{-2}])$ is increasing on $(0, 1/4]$, since its derivative is $\log_2(1+x^{-2}) + \log_2 e\,(1 - 2/(1+x^2)) \ge \log_2 17 - \log_2 e > 0$ there. □

*Proof of Theorem 1.* Let $k = \lfloor 1/\eta \rfloor + 1$, so $k\eta > 1$, and let $L \to \infty$ with $D \le 2^{(1-\eta)L}$. Then $Da \le 2^{-\eta L}$, so $L(D-1)a \to 0$, $\theta \to 1$, $k(D-1)a < 2$ and $da \le Da \le 1/4$ for $L$ large, uniformly in $d \le D$. Next $d\gamma_k \le Da\, kL^k \le kL^k 2^{-\eta L} \to 0$, and

$$\kappa \le \frac{2L^k 2^{\binom{k}{2}} (Da)^k}{a\theta} \le \frac{2L^k 2^{\binom{k}{2}}}{\theta}\, 2^{(1-k\eta)L} \to 0.$$

In the third term, $x \mapsto x\log_2(e[3+c/x])$ is increasing, $\varepsilon \le kL^k 2^{-\eta L}$, and $c = 8/(da\theta) \le 16 \cdot 2^L$ for $L$ large, so the term is $O(kL^{k+1}2^{-\eta L})$. The last term is at most $2^{-\eta L}\log_2(e[1+4^{\eta L}]) = O(L 2^{-\eta L})$. None of these bounds depends on $n$, on the number of terms or on $i$. □

*Remark* 7. The two error sources pull $k$ in opposite directions. Since $\gamma_k \le kL^{k-1}a$, the weight of $h$ off $\Omega_k$ is small once $DkL^{k-1} \ll 2^L$, which favours small $k$; the weight of $N$ on $\Omega_k$ is small once $L^k D^k \ll 2^{(k-1)L}$ up to factors depending on $k$, which favours large $k$. With $D = 2^{\beta L}$, $\beta < 1$, both hold for $k > 1/(1-\beta)$, and this is where the hypothesis $D \le 2^{(1-\eta)L}$ comes from.

## 4. Numerical checks

All values below were computed with exact Walsh–Hadamard transforms. The script `check_lin.py` in the code repository evaluates the four inequalities of Lemmas 3–5 and Theorem 6 at every coordinate and every admissible $k$ for 17 linear DNFs: stars of $D$ terms through one variable, the rows and columns of a $4 \times 4$ array, and twelve random linear DNFs with $w \in \{4, 5\}$, $D \in \{2, 3, 4\}$ and 16 to 18 variables. All 784 (coordinate, $k$) pairs satisfy all four inequalities, and the left side of Theorem 6 is never more than 0.14 of the right side; at these widths the bound is far from tight.

Table 1 shows the ratio approaching 2 as the width grows, for the star of $D$ terms of width $w$ through a common variable, at the centre ($d = D$) and at another variable ($d = 1$). For the rows and columns of an $m \times m$ array the ratio is 2.629 at $m = 3$ and 3.245 at $m = 4$.

| $D$ | $w$ | $n$ | Inf centre | $E$/Inf centre | Inf other | $E$/Inf other |
|---|---|---|---|---|---|---|
| 2 | 4 | 7 | 0.2344 | 2.685 | 0.1094 | 2.909 |
| 2 | 6 | 11 | 0.0615 | 2.450 | 0.0303 | 2.351 |
| 2 | 8 | 15 | 0.0156 | 2.178 | 0.0078 | 2.118 |
| 2 | 10 | 19 | 0.0039 | 2.060 | 0.0019 | 2.037 |
| 3 | 5 | 13 | 0.1760 | 2.692 | 0.0549 | 2.666 |
| 3 | 7 | 19 | 0.0461 | 2.378 | 0.0151 | 2.225 |
| 4 | 5 | 17 | 0.2275 | 2.690 | 0.0515 | 2.728 |

TABLE 1. $E_i/\mathrm{Inf}_i$ for the star of $D$ terms of width $w$ on $n = 1 + D(w-1)$ variables.

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This work was done by multiple AI agents, Claude (Anthropic) and Codex (OpenAI), connected to Math Brain [16] and working together inside the Brain framework [18], a shared, persistent memory in which the agents record claims, proofs, failed attempts and methods, retrieve one another's results, coordinate who works on what, and audit each other's work. For more on the details behind the Brain and Math Brain, see the Math Brain paper [17], the Math Brain repository [16] and the Brain framework repository [18].

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## Declaration of competing interest

The author holds equity in Heights Labs and in Convex Nexus Capital and has a management role at Convex Nexus Capital. These entities are included in the funding declaration above.

## Declaration of generative AI and AI-assisted technologies

The paper comes out of Math Brain, an ongoing project on the use of AI systems for research mathematics, and its results were developed with the Brain [18] and Math Brain [16]. In November 2026 a Claude Fable agent (Anthropic) and Codex agents running OpenAI's GPT-6.1, connected to the Brain and working on the same problem, studied erasure entropies of DNFs; a Codex agent found the theorem for sparse linear DNFs and its proof. Claude Opus 5.5 (Anthropic) rederived each proof, wrote the numerical checks, prepared the manuscript and audited it. These checks are not peer review. The author directed the work and is responsible for the content of the paper and for decisions concerning its dissemination.

## Code and data availability

Code and outputs: `https://github.com/yspennstate/fei-sparse-linear-dnf`.

The paper is part of the Math Brain project, whose public repository is `https://github.com/yspennstate/math-brain-public` [16].

## Appendix A. Notation

Symbols are listed roughly in the order in which they first appear, and the last column gives that place. The letters $c$, $k$ and $S$ are reused, and each meaning has its own row; $D$ is the degree bound, and $D_j$, $D_J$ are derivatives.

| Symbol | Meaning | First appears |
|---|---|---|
| $f$ | a Boolean function $\{0,1\}^n \to \{-1,1\}$; for a DNF $F$, $f = 1 - 2F$ | Section 1 |
| $\hat{f}(S)$, $\chi_S$ | the Fourier coefficient of $f$ at $S$, and the character $(-1)^{\sum_{j\in S} x_j}$ | Section 1 |
| $U$ | a random subset of $[n]$ with $\Pr[U = S] = \hat{f}(S)^2$ | Section 1 |
| $H(f)$ | the Fourier entropy $H(U)$ | Section 1 |
| $\mathrm{Inf}_i(f)$, $I(f)$ | the influence $\Pr[i \in U]$ of $i$, and the total influence $\mathbb{E}\lvert U\rvert$ | Section 1 |
| $C$ | an absolute constant, as in $H(f) \le C\, I(f)$ | Section 1 |
| $k$ | the number of times a decision tree or DNF may read a variable, in read-$k$ | Section 1 |
| $E_i(f)$ | the erasure entropy $H(\mathbf{1}[i \in U] \mid U \setminus \{i\})$ | Section 1 |
| $c$ | an absolute constant, as in $E_i(f) \le c\,\mathrm{Inf}_i(f)$ | Section 1 |
| $h_2$ | the binary entropy | Section 1 |
| $F$ | a monotone DNF on $\{0,1\}^n$ | Section 1 |
| $w$, $D$ | the width of every term, and the largest number of terms through one variable | Section 1 |
| $m$ | the side of the $m \times m$ array whose rows and columns are the terms | Section 1 |
| $\eta$ | a fixed number in $(0,1)$, as in $D \le 2^{(1-\eta)(w-1)}$ | Theorem 1 |
| $k$ | the number of variables of an AND, less one, in the comparison with a single AND | Section 1 |
| $\Phi(a,b)$ | $\ell(a) + \ell(b) - \ell(a+b)$, with $\ell$ as below | Section 1 |
| $g$, $h$ | the even and odd parts $\frac{1}{2}(f^{(0)} \pm f^{(1)})$ of $f$ in $x_i$ | Section 1 |
| $N$ | the indicator that $F$ vanishes at $x_i = 0$ (for a linear DNF, that every background term fails) | Section 1 |
| $\Omega$ | a family of non-empty sets on which $h$ has most of its weight and $N$ little | Section 1 |
| $\Omega_k$ | the sets containing at least $k$ mates of some term through $i$ | Section 1 |
| $f^{(b)}$ | the restriction $x_i = b$ | Section 2 |
| $T$ | a subset of $[n] \setminus \{i\}$ | Section 1 |
| $\delta$ | the influence $\mathrm{Inf}_i(f) = \mathbb{E}h^2$ | Section 2 |
| $p(T)$, $q(T)$ | $\hat{g}(T)^2$ and $\hat{h}(T)^2$ | Section 2 |
| $\ell(x)$ | $-x \log_2 x$ | Section 2 |
| $M$ | the indicator that $F$ vanishes at $x_i = 1$ | Section 2 |
| $\varepsilon$, $\kappa$ | the weight of $h$ off $\Omega$ and the weight of $N$ on $\Omega$, as fractions of $\delta$ | Proposition 2 |
| $P$, $Q$ | the sums of $p$ and $q$ over the non-empty sets outside $\Omega$, in the proof | Proposition 2 |
| $c$ | $8\mathrm{Var}N + 3\varepsilon\delta$, in the proof | Proposition 2 |

| Symbol | Meaning | First appears |
|---|---|---|
| $L$, $a$ | $w-1$ and $2^{-L}$ | Section 3 |
| $d$, $t_1, \dots, t_d$ | the number of terms through $i$, and those terms | Section 3 |
| mates, $A_s$ | the $L$ variables of $t_s$ other than $i$, and the indicator that all of them equal 1 | Section 3 |
| background term | a term that does not contain $i$ | Section 3 |
| $A$ | $\mathbb{E}N$ | Section 3 |
| $\theta$ | $1-L(D-1)a-(d-1)a/2$ | Lemma 3 |
| $R_s$, $V_s$ | the indicator that every background term disjoint from the mates of $t_s$ fails, and the event that one residual conjunction of the others equals 1 | Lemma 3 |
| $k$ | the number of mates that puts a set in $\Omega_k$, $1 \le k \le L$ | Section 3 |
| $\gamma_k$ | $a\sum_{j<k}\binom{L}{j}$ | Section 3 |
| $\beta$ | the exponent in $D = 2^{\beta L}$, $\beta < 1$, in the closing remark | Section 3 |
| $h_s$, $\Psi_s$ | $NA_s\prod_{u<s}(1-A_u)$, and the same product with the mates of $t_s$ set to 1 | Lemma 4 |
| $S$, $V$ | a set of mates of $t_s$, and a set disjoint from them, in the proof | Lemma 4 |
| $\alpha_k$ | the constant bounding the weight of $N$ on $\Omega_k$ | Lemma 5 |
| $J$ | a set of $k$ mates of one term through $i$ | Lemma 5 |
| $D_j$, $D_J$ | the derivative $\frac{1}{2}(N\|_{x_j=0} - N\|_{x_j=1})$ in $x_j$, and the composition of these over $j \in J$ | Lemma 5 |
| $R'$, $Q_j$ | the indicator that every background term avoiding $J$ fails, and the OR of the residual conjunctions of the background terms through $j$ | Lemma 5 |

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